Prospective Q&A: Continuous-Time RL for Portfolio Optimization

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Questions marked **[Sabzikar]** are most likely to come from Dr. Sabzikar given his background in stochastic processes and probability theory. Questions marked **[Caution]** are ones where you should acknowledge a genuine limitation honestly rather than overselling the answer.

Part I: Questions Likely from Dr. Sabzikar

Q1: Boundary Classification of the Wonham Filter [Sabzikar]

Q: The paper claims $p = 0$ and $p = 1$ are entrance boundaries for the belief state process. Can you walk through why, using Feller's boundary classification, and what the mathematical consequence is for the PDE system?

A: Feller's boundary classification applies to one-dimensional diffusions $dY_t = b(Y_t)\,dt + \sigma(Y_t)\,dW_t$. For the belief state on $(0, 1)$:

$$dp_t = [-(\lambda_1 + \lambda_2)p_t + \lambda_2]\,dt + (\rho_1 - \rho_2)p_t(1 - p_t)\,d\hat{B}_t$$

the diffusion coefficient is $\sigma(p) = |\rho_1 - \rho_2|p(1 - p)$, which vanishes at both endpoints. Under Feller's classification, a boundary is an **entrance boundary** if the process cannot be started there but immediately enters the interior — formally verified by checking the integrability conditions on the scale function and speed measure near the boundary. Here, the drift pushes away from both endpoints (positive near 0, negative near 1) while the diffusion vanishes faster than the drift. Both endpoints satisfy the entrance boundary conditions.

Consequence for the PDE: Since a process started in $(0, 1)$ never reaches 0 or 1, the PDEs for f and g on $[0, T] \times (0, 1)$ do not require boundary conditions at the endpoints. Well-posedness follows from Theorem 10 of Krylov (2008) and the comparison principle in the paper's Lemma 1.

Note on the numerics: The finite-difference scheme (Appendix B) does need artificial boundary conditions because the grid has finite endpoints. The paper uses the Wang-Zhou (2020) single-regime formulas there, which are exact at $p = 0$ (certain bear) and $p = 1$ (certain bull).

Likely follow-up: "How does the comparison principle apply when the diffusion degenerates at the boundary?"

The proof uses localization: define stopping times $\tau_n = \inf\{t : p_t \notin (1/n, 1 - 1/n)\}$. Since the endpoints are entrance boundaries, $\tau_n \to \infty$ a.s. Establish the comparison on $[0, T \wedge \tau_n]$ where the diffusion is nondegenerate, then pass to the limit using dominated convergence.

Q2: Innovation Process and Measurability [Sabzikar]

Q: You write the innovation process as $d\hat{B}_t = [d \ln S_t - (\hat{\mu}_t - \sigma^2/2)\,dt]/\sigma$. For this to be a standard Brownian motion with respect to $\mathcal{F}_t^S$, which theorem are you invoking, and what are its conditions?

A: This is the **Fujisaki-Kallianpur-Kunita (FKK) theorem**, cited as Kallianpur (2013), Theorem 8.1.2. The conditions are: (1) the signal I_t is a finite-state CTMC with bounded transition rates; (2) the observation S_t has drift depending on I_t through $\mu_t = \mu_{I_t}$ with additive Brownian noise; (3) the filtered drift $\hat{\mu}_t = \mathbb{E}[\mu_t \mid \mathcal{F}_t^S]$ is $\mathcal{F}_t^S$-adapted and square-integrable.

Under these conditions, $\hat{B}_t$ is a standard $(\mathcal{F}_t^S)$-Brownian motion. It drives both the Wonham filter SDE and the observable wealth dynamics $dX_t = \sigma u_t(\hat{\rho}_t\,dt + d\hat{B}_t)$. Since $\hat{B}_t$ is constructed entirely from observables, the wealth dynamics are fully adapted to $\mathcal{F}_t^S$.

[Caution]: The paper cites Kallianpur (2013) but does not reproduce the proof. If pressed for more detail, acknowledge that the paper treats this as a standard result from nonlinear filtering theory.

Q3: Does the Separation Principle Apply to the Entropy-Regularized Problem? [Sabzikar]

Q: The separation principle you cite is from Xiong and Zhou (2007), which covers the standard MV problem. Wu and Li solve an entropy-regularized version. Does the separation principle still apply after adding the entropy term?

A: Yes, and this is handled in Section 2.3 of the paper, though implicitly. The exploratory wealth dynamics:

$$d\tilde{X}_t^{\boldsymbol{\pi}} = \sigma\hat{\rho}_t\bar{u}_t\,dt + \sigma\bar{u}_t\,d\hat{B}_t + \sigma\sqrt{\overline{u^2}_t - \bar{u}_t^2}\,d\bar{B}_t$$

are adapted to the expanded filtration $\bar{\mathcal{F}}_t = \mathcal{F}_t^S \vee \sigma(Z_s, s \le t)$ where Z_s are the randomization variables. The entropy objective $\mathcal{H}(\boldsymbol{\pi}(\cdot \mid t, X_t, p_t))$ depends on p_t, which is $\mathcal{F}_t^S$-adapted. So the full objective is measurable with respect to the observable filtration.

The separation works because the stochastic policy $\boldsymbol{\pi}$ is a function of (t, X_t, p_t) — all observable — and the entropy regularizer adds a running cost that also depends only on observables. The belief state p_t serves as the sufficient statistic for the unobservable regime.

Likely follow-up: "Doesn't the randomization noise $\bar{B}_t$ create a measurability issue?"

No. Jia and Zhou (2022b) show the sample state process $\{X_t^\pi\}$ is identical in law to the exploratory state process $\{\tilde{X}_t^\pi\}$. The $\bar{B}_t$ noise is averaged out in the expectation. The value function is defined under the expanded measure $\bar{\mathbb{P}}$, which agrees with $\mathbb{P}$ when restricted to $\mathcal{F}_T^S$.

Q4: Well-Posedness of the f PDE [Sabzikar]

Q: The diffusion coefficient in the f PDE is $(\rho_1 - \rho_2)^2 p^2 (1-p)^2$ — a fourth-order polynomial in p that vanishes at the endpoints. Standard parabolic PDE theory requires uniform ellipticity. How do you establish well-posedness here?

A: On any compact interval $[\varepsilon, 1-\varepsilon]$ for $\varepsilon \in (0, 1/2)$, the diffusion coefficient is bounded below by $(\rho_1 - \rho_2)^2\varepsilon^2(1-\varepsilon)^2 > 0$, so the PDE is uniformly parabolic on compact subsets. Global well-posedness on $(0,1)$ follows from three steps:

1. Krylov (2008), Theorem 10 (Chapter 2) gives existence and uniqueness of a smooth solution on $[0,T) \times [\varepsilon, 1-\varepsilon]$ for each ε.
2. The comparison principle establishes that solutions agree on overlapping compact sets, yielding a unique global solution.
3. Boundedness of f on $[0,T] \times (0,1)$ follows from the bounds $1 \le f(t,p) \le e^{\bar{\rho}^2(T-t)}$, proved by constructing explicit super- and sub-solutions.

The entrance boundary behavior is essential: because p_t never hits 0 or 1, you never need to evaluate f at the degenerate points in the control problem.

Q5: The v_{xp} Cross-Derivative — Where Does It Come From? [Sabzikar]

Q: The HJB equation has a cross-derivative term $(\rho_1 - \rho_2)p(1-p)v_{xp}$. Can you explicitly show via Itô's formula where this comes from?

A: Apply Itô's formula to $v(t, X_t, p_t)$. The relevant cross-variation term is:

$$dX_t \cdot dp_t = \sigma u_t\,d\hat{B}_t \cdot (\rho_1 - \rho_2)p_t(1-p_t)\,d\hat{B}_t = \sigma u_t(\rho_1 - \rho_2)p_t(1-p_t)\,dt$$

because $d\hat{B}_t \cdot d\hat{B}_t = dt$. The exploration noise $d\bar{B}_t$ is independent of $d\hat{B}_t$, so $d\bar{B}_t \cdot dp_t = 0$.

This cross-variation produces the v_{xp} term in the HJB. When you complete the square over u to find the optimal $\boldsymbol{\pi}$, this term enters the mean of the Gaussian policy — it is the hedging term in Theorem 2:

$$m^* = \underbrace{-\frac{x-\omega}{\sigma}\hat{\rho}}_{\text{myopic}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1-\rho_2)p(1-p)\frac{f_p}{f}}_{\text{hedging from } v_{xp}}$$

In the GBM case (Wang and Zhou 2020), there is no p_t, no cross-variation, no v_{xp} term, and no hedging component. The hedging term is the direct consequence of regime switching.

Q6: Verification Theorem Conditions [Sabzikar]

Q: For the verification theorem (Theorem 2) to hold, you need to verify several conditions. What are they, and is there a gap in the proof?

A: The verification requires:

1. **Admissibility of $\boldsymbol{\pi}^*$:** Four conditions in the paper's definition. The key ones are: (ii) unique strong solution to the exploratory SDE under $\boldsymbol{\pi}^*$ — follows from linear growth and Lipschitz conditions on the drift, which hold because f_p/f is uniformly bounded (Theorem 1); (iii) and (iv) square-integrability conditions — follow from a moment bound on $\tilde{X}^*$ (equation 42 in the paper) and boundedness of f.
2. **Inequality $v(t,x,p) \leq V^\pi(t,x,p)$ for all admissible $\boldsymbol{\pi}$:** Apply Itô's formula to $v(t, \tilde{X}_t^\pi, p_t)$, use the HJB inequality, and take expectations. Stochastic integrals are martingales by square-integrability up to $T \wedge \tau_n$; dominated convergence passes the result to $t = T$.
3. **Equality for $\boldsymbol{\pi}^*$:** The HJB holds with equality at $\boldsymbol{\pi}^*$ by construction.

[Caution]: The proof relies on Krylov (2008) for regularity ($v \in C^{1,2}$ on compact subsets) and Karatzas and Shreve (1991) for the moment bound. It is rigorous but not self-contained — it invokes external results. If Dr. Sabzikar asks whether the proof is fully self-contained, the honest answer is no.

Q7: Martingale Loss — Uniqueness of Minimizer [Sabzikar]

Q: You claim the martingale loss $\mathcal{ML}(\theta;\phi)$ has a unique minimizer at the true value function parameters. Why doesn't the polynomial parametrization introduce multiple local minima?

A: The uniqueness result is from Jia and Zhou (2022a), Theorem 3.2. The martingale loss for a fixed ϕ is:

$$\mathcal{ML}(\theta;\phi) = \tfrac{1}{2}\mathbb{E}\int_0^T \left| M_T^{\boldsymbol{\pi}^\phi} - M_t^\theta \right|^2 dt$$

Uniqueness requires strict convexity in θ. The gradients $\partial V^\theta/\partial\theta_{i,j} = -(x-\omega)^2 p^i (T-t)^j / f^\theta$ and $\partial V^\theta/\partial\tilde{\theta}_{i,j} = p^i(T-t)^j$ are linearly independent as functions of (t,p) — polynomial basis functions are linearly independent. This ensures the Hessian of the martingale loss is positive definite, giving a unique global minimizer.

Practical caveat: The polynomial with $m = 2$ introduces approximation error — the true $f(t,p)$ is not exactly of the form exp(quadratic in $p, T-t$). The minimizer θ^* is the best polynomial approximation in the metric induced by the martingale loss. The 1.03% MAE confirms the approximation is tight.

Part II: Questions from the General PhD Audience

Q8: Why Not Just Use Deep RL (DDPG, SAC)?

Q: DDPG and SAC handle continuous state and action spaces. Why use a problem-specific polynomial parametrization?

A: The paper benchmarks against DDPG and SAC in related work (Wang and Zhou 2020, Huang et al. 2022) and the polynomial consistently wins for three reasons:

1. **Parameter efficiency.** The polynomial has 12 parameters total; a typical neural network actor-critic has thousands. 200,000 training episodes is enough for the polynomial to converge reliably but may be insufficient for a neural network to explore the parameter space adequately.
2. **Structural inductive bias.** The paper derives analytically that the value function is quadratic in x with coefficient $1/f^{\theta}(t,p)$, and the policy is Gaussian with mean proportional to $(x-\omega)$. The polynomial encodes this as a prior. A neural network has to rediscover this structure from data.
3. **Training stability.** DDPG and SAC are known to produce unstable training curves in this problem class. The polynomial's constrained structure avoids this.

The tradeoff: this only works because the solution structure is known analytically. For a problem without such structure, neural networks would be necessary.

Q9: What Happens if Λ Is Too Large or Too Small?

Q: You set $\Lambda = 1$ throughout with no sensitivity analysis. What are the consequences of misspecifying Λ?

A: The policy variance is $f(t,p)\Lambda/(2\sigma^2)$. Larger Λ means more exploration — wider sampling of investment amounts — which helps the critic generate diverse trajectories but increases trajectory noise for the actor. As $\Lambda \to 0$, the policy collapses to deterministic and training degenerates (no exploration). As $\Lambda \to \infty$, the policy randomizes investment amounts so heavily that no useful signal accumulates.

The paper uses $\Lambda = 1$, borrowed from Wang and Zhou (2020) where it worked well for the GBM case. No sensitivity analysis is provided, which is a genuine limitation. The empirical convergence in Fig. 5 is circumstantial evidence that $\Lambda = 1$ is in a reasonable range, but the optimal Λ is likely problem-dependent.

Q10: The Precommitted Strategy — Doesn't Time Inconsistency Undermine Practical Usefulness?

Q: The precommitted strategy is optimal at $t = 0$ but an investor at $t > 0$ would want to deviate. Doesn't this limit practical applicability?

A: This is a real issue and the paper is explicit about it. Two responses:

Theoretical: The equilibrium (time-consistent) approach for continuous-time MV was developed by Dai et al. (2023) for the GBM case. Extending to the regime-switching partial-information case is listed as Open Problem 2. The precommitted strategy is a deliberate scoping decision, not an oversight.

Practical: In delegated investment (a fund manager following a mandate), precommitment is natural — the manager commits to a strategy at $t = 0$ and executes it. The time inconsistency is not a practical problem in this context. The empirical results also show the precommitted policy executes well: terminal wealth mean is close to the target $z = 8$ across all experiments.

Q11: Is the Two-Regime Model Credible?

Q: Financial returns have fat tails, volatility clustering, and potentially more than two regime types. Is two-regime GBM adequate for long-horizon investment?

A: For long-horizon investment (10-year horizon), the dominant empirical regularity is the bull-bear cycle, which the two-regime model captures parsimoniously. Guidolin and Timmermann (2007, 2008) document that two-regime models describe long-horizon return dynamics well. The calibrated parameters ($\mu_1 = 0.25$, $\mu_2 = -0.73$, average bull duration ≈ 2.8 years) are economically reasonable, and the empirical out-of-sample performance on 30 years of S&P 500 data provides at least circumstantial validation.

The constant volatility assumption is more limiting — bear markets empirically have higher volatility — but regime-dependent volatility would fundamentally change the filtering problem (the quadratic variation of $\ln S_t$ would reveal the regime). More regimes would require a higher-dimensional belief state $p_t \in \mathbb{R}^{K-1}$ and a corresponding increase in the PDE dimension.

Q12: How Does Exploration Wind Down Over Time?

Q: Standard RL uses a decay schedule for ϵ-greedy or temperature. How does POEMV handle exploration decay, and is it automatic?

A: It is automatic and built into the solution structure. The policy variance is $f(t,p)\Lambda/(2\sigma^2)$, and $f(t,p)$ is proved to be decreasing in t (Theorem 1). So exploration naturally winds down as $t \to T$ without any manual schedule. The intuition is correct: near the terminal time, committing to a more deterministic allocation matters more than continued exploration. The parametrized f^θ inherits this monotonicity approximately, so the learned policy also reduces exploration over the investment horizon.

Q13: Why Does the Self-Correcting ω Update Work?

Q: You update $\omega \leftarrow \omega - \eta_\omega(\bar{X}_T - z)$. Why does this converge to ω^*?

A: Define $h(\omega) = \mathbb{E}[X_T^{u^*(\omega)}] - z$. The optimal ω^* satisfies $h(\omega^*) = 0$. The update rule is a **Robbins-Monro stochastic approximation** algorithm for solving $h(\omega) = 0$, using $\bar{X}_T - z$ as a noisy estimate of $h(\omega)$.

Convergence requires $h(\omega)$ to be monotone decreasing in ω. This holds because a higher ω increases $(\omega - x)$ in the mean policy $m^* = -(x-\omega)\gamma^*(t,p)$, leading to larger positions and higher expected terminal wealth. With h decreasing and Robbins-Monro conditions satisfied, $\omega^{(n)} \to \omega^*$.

The every-s-episode update (using average terminal wealth over $s = 10$ episodes) reduces noise in $\hat{h}$, improving convergence stability.

Q14: Why Does Monthly Rebalancing Hurt the Classical Approach So Much More?

Q: POEMV variance is 0.284 under monthly rebalancing; classical is 10.081. That's a 35x gap. Why?

A: Two compounding effects:

1. **Mean blur compounds over the rebalancing window.** The classical policy uses inaccurate $\hat{\mu}_i$, creating a fixed allocation bias. Under daily rebalancing this is corrected continuously. Under monthly rebalancing, the biased allocation persists for 20 trading days at a time, compounding the error multiplicatively.

2. **Regime transitions between rebalancing dates.** The market can switch regimes during the 20-day window. The classical policy cannot respond because it doesn't condition on p_t. POEMV conditions on the belief state at each monthly rebalancing date and adjusts accordingly.

The 10.081 variance reflects a small number of catastrophic paths where estimated parameters implied aggressive allocation into a bear market for extended periods. POEMV's regime-awareness prevents these tail events.

Q15: Is the Universal Training Trick In-Sample?

Q: You train on 30 $\beta \approx 1$ stocks and test on those same 30 stocks plus the index. Isn't the test set the training set?

A: Partially yes, which is a legitimate concern. The defense is: the policy being learned maps (t, X_t, p_t) to an action — it is not path-specific. The same policy applies to any asset with similar statistical properties. Training on 30 stocks addresses data scarcity (one index path gives ~5,000 observations; 30 stock paths give ~150,000). The train-test split is temporal: training uses 1992–2012 data, testing uses 2012–2022 realizations. The policy is not fitted to the specific 2012–2022 paths.

A cleaner design would train on different stocks from the test stocks. The paper does not do this — a limitation worth acknowledging. However, since the policy is state-based rather than path-memorizing, the overfitting risk is lower than it would be for a path-interpolating model.

Q16: How Long Does Training Take?

Q: 200,000 episodes of 2,520 steps each is 504 million steps. Is this feasible computationally?

A: The paper does not report wall-clock times. However, the computation per step is lightweight: update the Wonham filter (a scalar SDE update), sample from a Gaussian, compute gradients for 12 parameters — all $O(1)$ scalar operations with no neural network backpropagation. The gradient update for θ involves a sum over $K = 2{,}520$ steps, so training is $O(MK) \approx 5 \times 10^8$ scalar operations total. On modern hardware this runs in minutes to a few hours — far faster than comparable neural network actor-critic approaches. The small parameter count is the key computational advantage.

Part III: Assumptions and Limitations

Q17: Constant Volatility

Q: Bear markets empirically have substantially higher volatility. How sensitive are results to the constant σ assumption?

A: Constant volatility is standard in this literature and is mathematically motivated: if σ were regime-dependent, the quadratic variation of $\ln S_t$ would reveal the regime, trivializing the filtering problem. With constant σ, only the drift is unobservable. The sensitivity to this assumption is genuinely unknown — the paper does not address it. If σ were regime-dependent, the separation principle derivation would need substantial revision since it relies on $dX_t = \sigma u_t(\hat{\rho}_t\, dt + d\hat{B}_t)$ having constant σ.

Q18: Transaction Costs

Q: Daily rebalancing must incur substantial transaction costs. Does the paper account for this?

A: No. The paper explicitly states (footnote 4) that the comparison ignores transaction costs. Monthly rebalancing results are partly motivated by this concern, and POEMV still substantially outperforms at monthly frequency. Incorporating transaction costs would require a non-smooth cost term in the HJB equation, fundamentally changing the structure of the solution. This is left as future work.

Q19: Is the Heuristic Estimation Circular?

Q: The heuristic method defines bull/bear markets using fixed thresholds and then estimates parameters within those segments. Isn't this circular?

A: There is mild circularity: the thresholds (19% decline, 24% advance) presuppose a two-regime structure. A fully principled approach would estimate regime structure and parameters jointly — which is what EM does, and as Table 2 shows, EM fails catastrophically at daily frequency. The defense of the heuristic is pragmatic: the thresholds are economically motivated rather than data-driven, and the resulting estimates are reasonable. More importantly, filter performance is validated directly in Fig. 4 — the belief state tracks regime transitions well despite imperfect parameter estimates, which is the empirical justification for the approach.

Q20: Does the Lack of a Convergence Guarantee Matter Practically?

Q: Is the missing end-to-end convergence guarantee just a theoretical nicety? The algorithm clearly converges in simulations.

A: It matters both practically and theoretically. Practically: the simulations show convergence for the specific parameters in Table 1 and specific learning rates. Without a guarantee, we cannot rule out pathological behavior for different parameters or market environments. A practitioner deploying POEMV in a different context has no theoretical assurance of convergence.

Theoretically: the three error sources — filter error, polynomial approximation error, stochastic gradient error — interact in ways that are not currently bounded. Filter error corrupts p_t, which corrupts the gradient estimates for ϕ, which means the converged ϕ^∞ may differ from ϕ^* even if the stochastic gradient step itself converges. Fig. 5b already shows this: with estimated parameters, ϕ_1 converges to ≈ 15 rather than the true 20.25. A convergence theorem would quantify exactly how large the filter error must be before the policy gap exceeds a given tolerance.

Part IV: Connections to Broader Literature

Q21: Why Not Standard POMDP Algorithms?

Q: This is a POMDP. Why not discretize time and use standard POMDP algorithms like point-based value iteration?

A: Three reasons to prefer the continuous-time formulation:

1. **Tractability.** Standard POMDP algorithms scale poorly with the belief space dimension. The continuous-time formulation with Wonham filtering gives a semi-analytical solution — a specific parametric form for the value function and policy — which makes learning efficient. Discrete-time algorithms would discretize both p and X, leading to combinatorial explosion.
2. **Solution structure.** The continuous-time formulation reveals the Gaussian optimal policy with myopic-hedging decomposition. This structure is invisible in discrete time and enables the 12-parameter polynomial to fit the solution precisely.
3. **No discretization bias.** The continuous-time formulation is exact. The discrete training episodes ($\Delta t = 1/252$) are an approximation of the exact dynamics, and the paper shows this approximation is tight.

Q22: What Does This Add Beyond Xiong and Zhou (2007)?

Q: Xiong and Zhou (2007) already solved the classical MV problem with partial information. What does this paper add?

A: Three things: (1) **Exploratory formulation.** Wu and Li add entropy regularization and solve the resulting exploratory problem under partial information. The cross-covariance $\mathrm{Cov}(\Delta X, \Delta p)$ makes this technically non-trivial — Xiong-Zhou has no exploratory component. (2) **Solution characterization.** The paper derives the structural form of the optimal value function (the ansatz) and establishes properties of f and g — bounds, monotonicity, boundedness of derivatives — that are needed for the algorithm. (3) **Practical algorithm.** POEMV learns the optimal policy from data without needing the true parameters for anything other than filtering. Xiong-Zhou's solution requires full knowledge of $(\mu_i, \lambda_i, \sigma)$, and as Table 2 shows, accurately estimating (μ_i, λ_i) at daily frequency is effectively impossible.

Q23: Why Wonham Filter and Not Kalman?

Q: The Kalman filter is the standard tool for filtering. Why use the Wonham filter?

A: The Kalman filter applies to **linear-Gaussian** state-space models. The regime process I_t is a finite-state CTMC, not a Gaussian process. The Wonham filter is the exact nonlinear filter for this specific structure (CTMC signal, Gaussian observation noise). The belief state $p_t \in (0, 1)$ is the natural sufficient statistic and its dynamics are given exactly by the Wonham equation — no linear approximation is involved.

Applying the Kalman filter here would require linearizing the observation model around the current p_t estimate (extended Kalman filter), introducing linearization error. The Wonham filter is exact and has no such error.

Quick Reference: One-Sentence Answers

Question	Short Answer
Why two regimes?	Standard parsimonious model for bull-bear cycles, calibrated to S&P 500, empirically sufficient at long horizons.
Why polynomial, not neural network?	Solution structure is known analytically — 12 parameters fit it precisely and train stably.
Why not EM for estimation?	Signal-to-noise too low at daily frequency; EM identifies regime switches from noise ($\hat{\lambda}_i \approx 128$).
Why does the martingale loss work?	Misspecification of V^θ creates nonzero drift in M_t^θ; minimizing the loss enforces the martingale property.
Where does the hedging term come from?	From $dX_t \cdot dp_t \neq 0$, which produces v_{xp} in the HJB — absent in the single-regime GBM case.
Is filtering error bounded?	Informally (Fig. 4 shows good tracking), but no formal bound is provided.
Does this generalize to multiple assets?	Yes — Appendix A of the paper extends to d risky assets with multivariate Gaussian policy.
What is the dominant source of suboptimality?	Filtering error from imprecise parameter estimation, not the learning algorithm.
Why is constant σ assumed?	Regime-dependent σ would reveal the regime via quadratic variation, changing the filtering problem entirely.
What's the key thing Wonham gives you over Kalman?	Wonham is exact for a CTMC signal; Kalman requires a linear-Gaussian model, which the regime process is not.